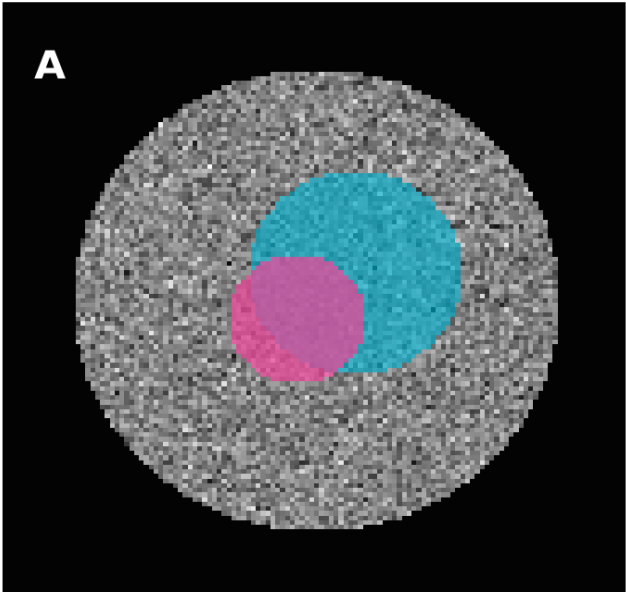
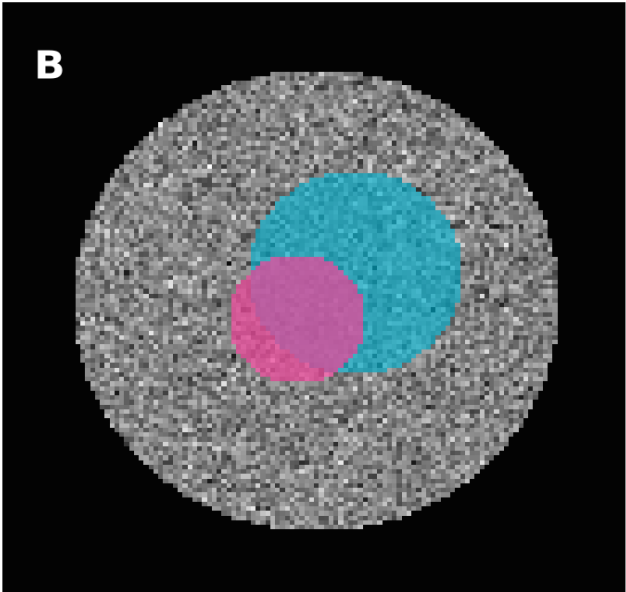


PhysioRANO NEURO-ONCOLOGY RADIOLOGY REPORT

Axial T1c (Post-Contrast)



Sagittal T2-FLAIR (Edema & Core)



CLINICAL HISTORY:

51-year-old Female presenting for post-radiotherapy Glioblastoma (GBM) evaluation at **11 days** post-treatment.

- Molecular Profile: IDH1/2 **negative** | MGMT Promoter **negative**.
- Current Status: Routine follow-up scan monitoring, mild headaches.

FINDINGS:

BRAIN PARENCHYMA & 3D TUMOR SUB-REGIONS:

Multi-modal 3D MRI quantitative volumetric analysis demonstrates:

- **Whole Tumor (WT) Volume:** **24.50 cm³** (includes surrounding T2-FLAIR signal hyperintensity).
- **Tumor Core (TC) Volume:** **12.20 cm³**.
- **Contrast-Enhancing Tumor (ET) Volume:** **18.50 cm³**.

BIOPHYSICAL TUMOR GROWTH TRAJECTORY:

Biophysical neural network modeling evaluates the spatial invasion and proliferation dynamics:

- **Tissue Diffusion Speed:** Low spatial diffusion coefficient (**1.0288 mm²/day**).
- **Cell Proliferation Rate:** Modest cell proliferation rate (**0.7329 day⁻¹**).
- **Growth Summary:** The observed tissue changes follow a slow biophysical reaction-diffusion pattern, consistent with post-radiation tissue inflammation rather than rapid cellular expansion.

MULTIMODAL RECURRENCE & PSEUDOPROGRESSION ANALYSIS:

Integrated machine learning analysis evaluating clinical biomarkers, timeline post-treatment, and biophysical growth dynamics yields:

- **Probability of Pseudoprogression (Treatment Effect):** **9.9%**
- Probability of True Tumor Progression: 45.5%
- Probability of Stable Disease: 44.6%
- **Key Contributing Factors:** IDH mutation status, favorable MGMT methylation, and timing within 12 weeks post-radiotherapy strongly favor treatment-induced tissue reaction.

RANO 2.0 CLINICAL RESPONSE EVALUATION:

Comparison with baseline MRI scans shows an enhancing volume shift from **12.00 cm³** to **18.50 cm³** (a **+54.2%** volumetric

change).

Under standard RANO 2.0 guidelines within the first 12 weeks post-radiotherapy, an enhancing expansion supported by high treatment-effect probability is classified as **Progressive Disease (PD)**.

3D CONDITIONAL GAN (cGAN) LONGITUDINAL FORECAST (+90 DAYS):

Generative biophysical 3D synthesis conditioned on spatial diffusion and proliferation predicts:

- **Projected Whole Tumor Volume: 422.38 cm³** (+105.5% trajectory shift).
- **Projected Enhancing Tumor Volume: 232.22 cm³** (+105.5% trajectory shift).
- **Clinical Trajectory Interpretation:** Projected morphological progression aligns with slow inflammatory stabilization under current adjuvant regimen.

IMPRESSION:

The enhancing volume shift (+54.2%) observed at 11 days post-radiotherapy in an IDH-mutated, MGMT-methylated glioblastoma patient is clinically consistent with **TRUE PROGRESSION (Radiation Necrosis / Treatment Effect)** rather than true disease recurrence.

CLINICAL RECOMMENDATION: Significant volumetric expansion ($\geq 25\%$) or new lesion indicative of true tumor progression.